



MEET SAFARSETU

VIHARAN 8.0

NUMBER PLATE
RECOGNITION WITH
DEHAZING, DERAINING
AND LEVERAGING
ESRGAN

DYNAMIC
TRAFFIC
LIGHTS

ACCIDENT
DETECTION
SYSTEM

OVERSPEED
DETECTION

TEAM CLASSY-CRUX
ISHANSH SHARMA (1ST YEAR)
ABHINAV RAJPUT (1ST YEAR)



PROBLEM STATEMENT-TRANSPORTATION

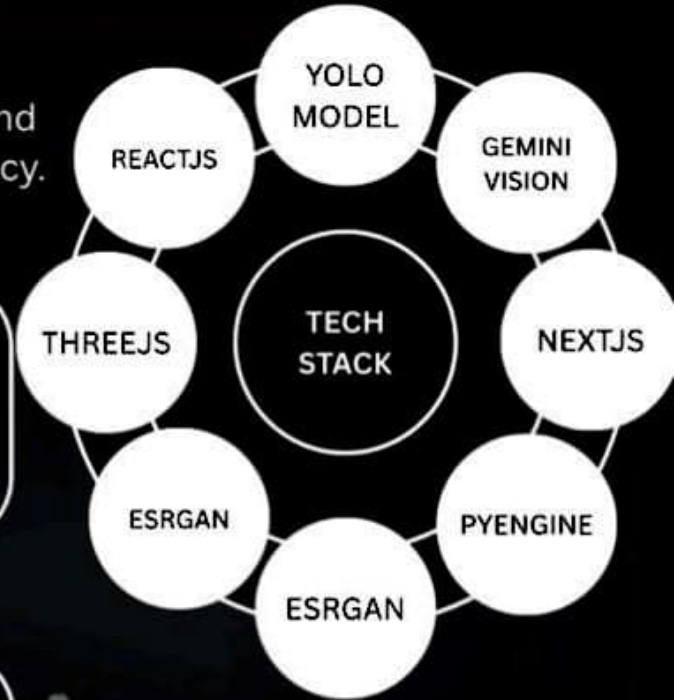


Urban traffic systems **lack real-time adaptability**, causing **congestion** and **delayed accident response**, **Poor visibility** further reduces ANPR accuracy. There's a need for an AI-driven solution to optimize traffic flow.

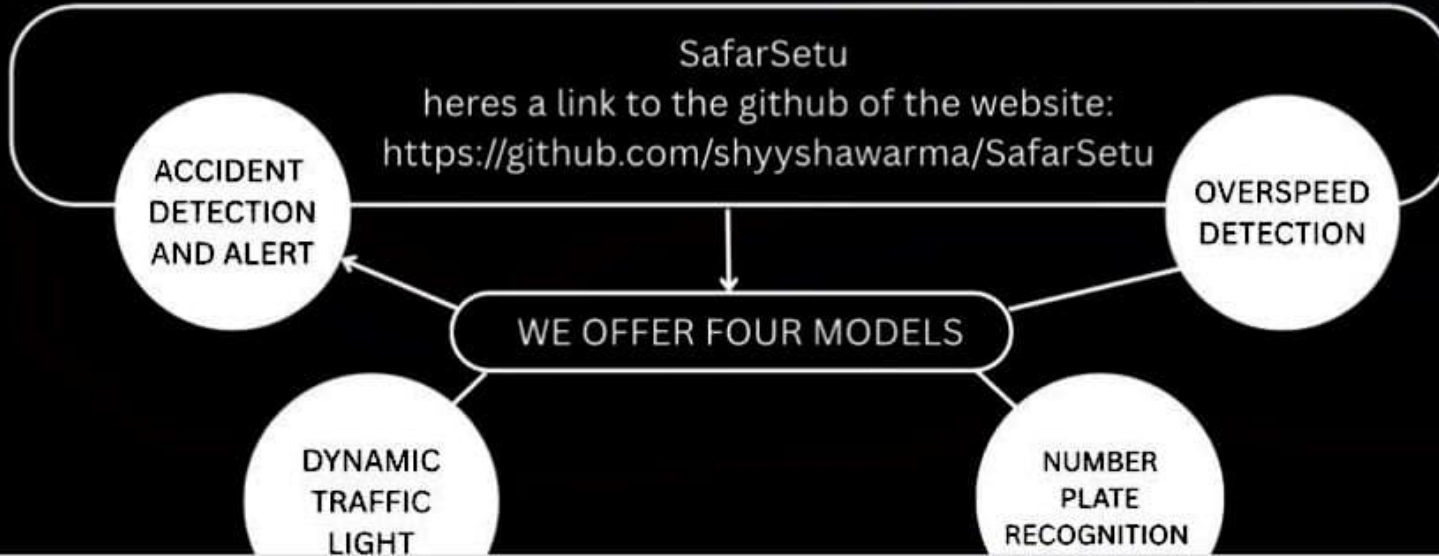


EXISTING SOLUTION

Cities rely on **static traffic systems** and **lack real-time data processing**. Without intelligent automation, AS accidents go unnoticed and speed violations are missed and poor ANPR techniques.



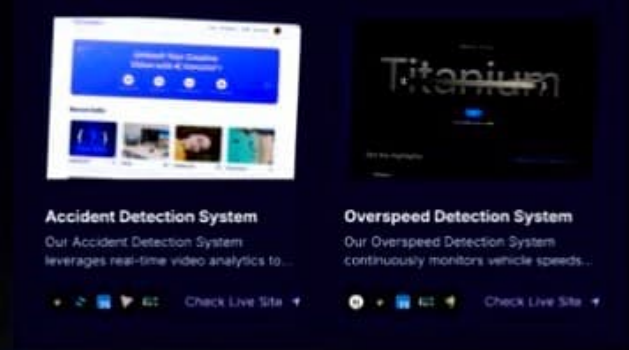
OUR SOLUTION



SAFARSETU PIPELINE AND WEB INTERFACE



We will provide complete package for Indian roads to move the traffic faster.



Dynamic Traffic Lights

Numberplate recognition with dehazing, deraining and ESRGAN

Accident detection system

Overspeed detection system



DYNAMIC TRAFFIC LIGHT MODEL



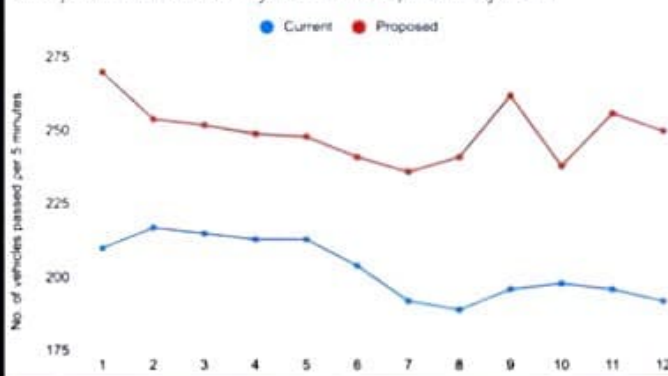
This leverages the use of yolo model to detect the traffic on a road and predict the average time required by the traffic to cross the intersection.

HERES A LINK TO THE GITHUB:

<https://github.com/shyyshawarma/Dynamic-traffic-light>

Results

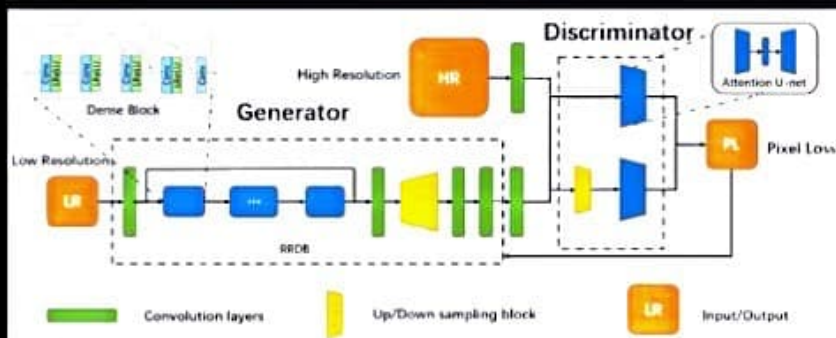
Comparison: Current System vs Proposed System



2435 vehicles passes through in the current system and 2997 in the proposed system.

Current System					Proposed Adaptive System					
Simulation No.	Lane 1	Lane 2	Lane 3	Lane 4	Total	Simulation No.	Lane 1	Lane 2	Lane 3	Total
1	87	76	51	9	223	1	111	88	42	241
2	76	73	47	9	205	2	105	82	46	233
3	80	73	33	28	214	3	110	86	46	242
4	76	71	38	27	212	4	98	73	38	209
5	77	66	44	38	225	5	85	88	42	215
6	76	72	37	27	212	6	77	87	37	201
7	69	73	36	16	194	7	73	82	46	201
8	67	66	33	28	194	8	77	82	46	205
9	49	81	16	28	174	9	80	66	46	192
10	57	71	46	35	209	10	75	82	37	194
11	52	71	38	46	207	11	110	105	38	253
12	69	71	28	38	206	12	76	87	42	205
2435					2997					

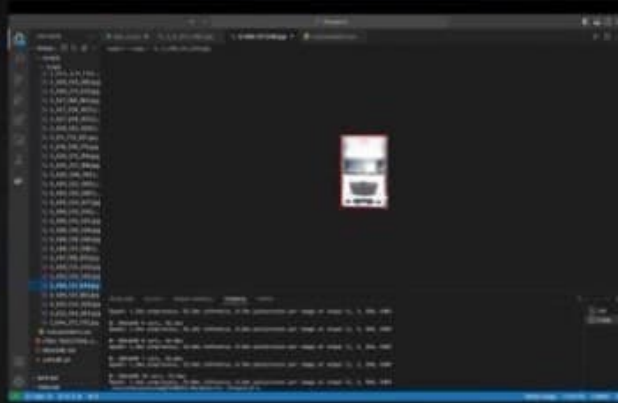
ANPR & ESRGAN MODEL



Detect number plate from the car and use Gemini vision or ESRGAN model to check the number plate.

We use techniques like dehazing and Deraining to get clear images. ESRGAN provides high resolution identification of image.

OVERSPEED DETECTION AND VISUAL LOGGING SYSTEM



Github link : <https://github.com/Ishansh-Sharma/Speed-Detector->

SMART ACCIDENT DETECTION

We will fine tune YOLO model to detect accidents on roads. We created a dataset using ROBOFLOW and then finetuned our model on a collab notebook.

FUTURE SCOPE

1. Implementing RL for dynamic traffic signals
2. Using Gemini vision instead of ESRGAN
3. Fine tuning yolo models for ambulance detection

Working

- Detection of different classes of vehicles using YOLO .
- Speed detection of vehicles using mathematical formula .
- Logging system capturing the frames and vehicles overspeeding .
- Vehicles id and speed also saved in a CSV file

